



AI SYSTEMS

SEMANTIC SEARCH

KOHLER 2026

# KOHLER AI Bathroom Designer & Planner

An AI planning engine that understands user preferences through semantic search and matches KOHLER products based on space, compatibility, and budget constraints.

## FOCUS AREA

AI-Based Bathroom Space  
Optimization

## CORE LOGIC

Constraint-Based Optimization

## OUTPUT

Verified Product Bundles

# Intelligent **KOHLER** Product Recommendation

Develop an AI-powered bathroom design assistant that understands customer requirements and generates personalized, space-aware, and budget-friendly KOHLER product bundles.

**A**

## **SPACE CONSTRAINTS**

AI designs may ignore fixture dimensions, clearances, and door space, producing impractical bathroom layouts.

**B**

## **PRODUCT MATCHING**

Finding products that match the customer's style, requirements, and space requires manual comparison across the catalog.

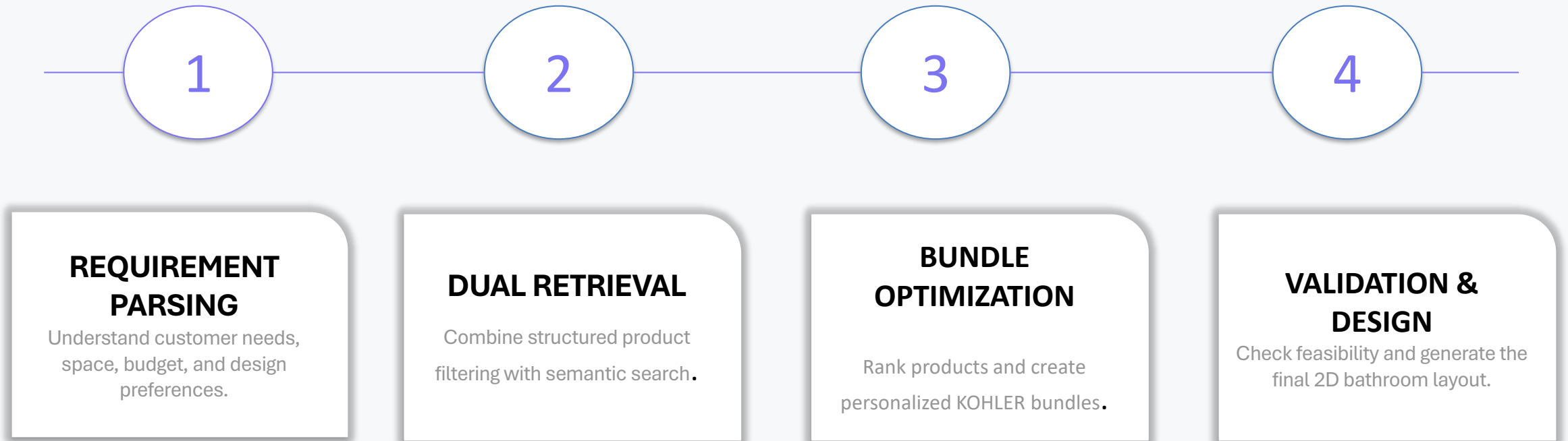
**C**

## **BUDGET CONSTRAINTS**

Selecting multiple bathroom products within a fixed budget is difficult without automated bundle-level optimization.

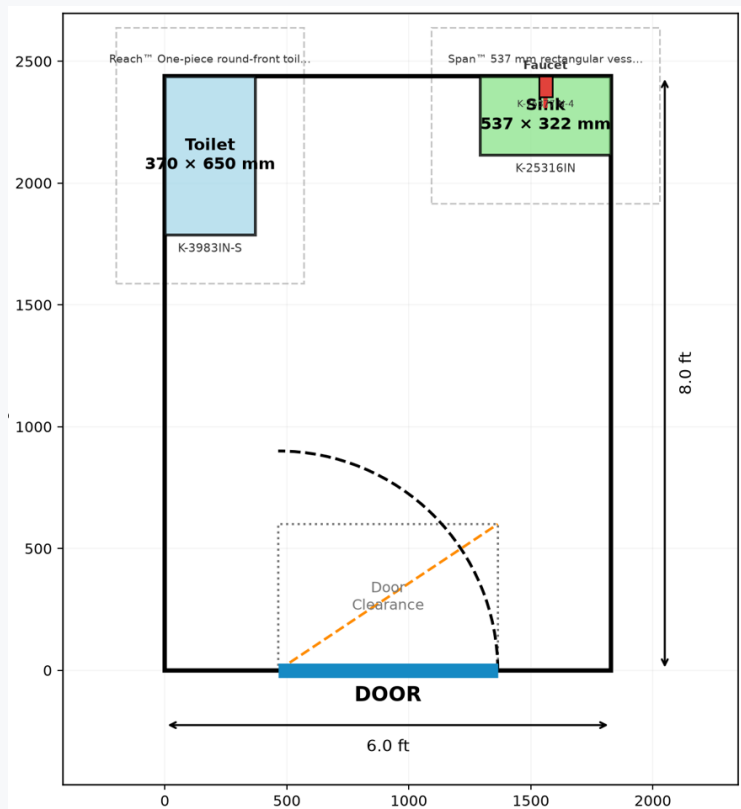
# HYBRID RAG ARCHITECTURE

Hybrid RAG combines structured product data with semantic search to find KOHLER products that match both customer constraints and design preferences.



# 2D Bathroom Layout

Generates a practical 2D bathroom layout by positioning fixtures according to available space, required clearances, and door movement.



## SPATIAL FEASIBILITY

Ensures fixtures fit within the available bathroom dimensions without overlapping or blocking movement.



## CLEARANCE AWARENESS

Maintains required clearance around fixtures and keeps door movement and access paths unobstructed



## VISUAL LAYOUT

Generates a clear 2D floor plan showing fixture placement, dimensions, door position, and clearance zones.